1. When configuring Network Address Translation (NAT) on a Cisco router, which command would you use to allow the translation of multiple inside local addresses to a single inside global address using PAT (Port Address Translation)?

Options:
A. `ip nat pool netmask`
B. `ip nat inside source list interface overload`
C. `ip nat outside source list interface overload`
D. `ip nat inside source static`

✅ **Correct Answer:**
B. `ip nat inside source list interface overload`

Explanation:
The command `ip nat inside source list interface overload` enables **PAT (Port Address Translation)**, allowing **multiple private (inside local) IP addresses to share a single public (inside global) IP address** by using different port numbers.

This is commonly used in home and enterprise networks to allow many devices to access the internet using one public IP address.

Example configuration:

```
access-list 1 permit 192.168.1.0 0.0.0.255
ip nat inside source list 1 interface GigabitEthernet0/0 overload
```

2. **Question:**
Which of the following statements about IPv6 addressing is true?

Options:
A. IPv6 employs an embedded IPv4 to IPv6 translation scheme using the **6to4 addressing format**.
B. IPv6 supports a **broadcast address** which allows communication to all nodes on a network.
C. IPv6 uses **link-local addresses, starting with fe80::/10, for communication within the same network segment.**
D. IPv6 addresses consist of **64 bits, composed of four 16-bit hexadecimal blocks.**

✅ **Correct Answer:**
C. IPv6 uses link-local addresses, starting with fe80::/10, for communication within the same network segment.

Explanation:
IPv6 automatically assigns **link-local addresses (fe80::/10)** to interfaces. These addresses are used for **communication within the same local network (LAN)** and are required for operations such as **Neighbor Discovery Protocol (NDP)**.

Why the others are incorrect:

- **A:** 6to4 is a **transition mechanism**, not a core addressing property of IPv6.
- **B:** IPv6 **does not use broadcast**; it uses **multicast and anycast instead**.
- **D:** IPv6 addresses are **128 bits**, written as **eight 16-bit hexadecimal blocks**, not 64 bits.

3. In a large campus network with multiple VLANs, which method optimizes inter-VLAN routing by leveraging Layer 3 switches?

Options:
A. Leveraging Layer 3 switches with SVIs (Switch Virtual Interfaces)
B. Implementing STP (Spanning Tree Protocol)
C. Using Layer 2 switches with VLAN trunking
D. Using traditional routers with subinterfaces

✅ **Correct Answer:**
A. Leveraging Layer 3 switches with SVIs (Switch Virtual Interfaces)

Explanation:
Layer 3 switches can perform **inter-VLAN routing internally using SVIs (Switch Virtual Interfaces)**. Each VLAN is assigned an SVI that acts as its default gateway, allowing routing to occur **directly on the switch hardware**, which is much faster than using an external router.

Example:

```
interface vlan 10
 ip address 192.168.10.1 255.255.255.0
 no shutdown
```

Why the others are incorrect:

- **B. STP:** Prevents **Layer 2 loops**, not used for routing between VLANs.
- **C. Layer 2 trunking:** Allows VLAN traffic between switches but **does not perform routing**.
- **D. Router with subinterfaces:** Known as **Router-on-a-Stick**, which works but is **less efficient than Layer 3 switching** for large campus networks.

4. In a network scenario, which is the primary reason that you would recommend using Network Address Translation 64 (NAT64) for IPv6-to-IPv4 connectivity?

Options:
A. To conservatively assign a large number of public IPv4 addresses for IPv6 clients.
B. To secure IPv4 networks from unauthorized IPv6 traffic entering through IPv6 tunnels.
C. To facilitate communication between IPv6-only clients and IPv4-only servers by translating the IPv6 packets to IPv4 packets.
D. To prioritize and manage bandwidth more efficiently across hybrid IPv4/IPv6 networks.

✅ **Correct Answer:**
C. To facilitate communication between IPv6-only clients and IPv4-only servers by translating the IPv6 packets to IPv4 packets.

Explanation:
NAT64 allows **IPv6-only devices** to communicate with **IPv4-only services** by translating IPv6 packets into IPv4 packets and vice versa. This is commonly used during the **IPv4 to IPv6 transition period**, where many networks are IPv6 but still need to access legacy IPv4 systems.

Why the others are incorrect:

- **A:** NAT64 does **not allocate additional IPv4 addresses**.
- **B:** NAT64 is **not primarily a security mechanism**.
- **D:** NAT64 is used for **protocol translation**, not bandwidth management.

5. What is the primary benefit of using 802.1X with dynamic VLAN assignment in a campus network?

Options:
A. It improves network performance by enabling traffic shaping based on VLAN membership.

B. It enhances network redundancy by allowing multiple VLANs to share the same physical switch ports.
C. It simplifies IP addressing by allowing clients to use the same IP address regardless of their location in the network.
D. It allows for both wired and wireless clients to be assigned to specific VLANs dynamically based on their authentication credentials.

✅ **Correct Answer:**
D. It allows for both wired and wireless clients to be assigned to specific VLANs dynamically based on their authentication credentials.

Explanation:
802.1X authentication uses a **RADIUS server** to verify user or device credentials. After successful authentication, the network can **dynamically assign the device to a specific VLAN** based on policies (user role, device type, security level).

This allows organizations to implement **role-based network access control**, improving **security and network segmentation** for both **wired and wireless clients**.

Why the others are incorrect:

- **A:** Traffic shaping is related to **QoS**, not 802.1X VLAN assignment.
- **B:** VLANs sharing switch ports is done using **trunking (802.1Q)**, not 802.1X.
- **C:** Dynamic VLANs **do not allow the same IP to be reused everywhere**; DHCP and subnetting still apply.

6. In a Cisco hierarchical network design, where would you most appropriately place a policy enforcement device such as an ACL-enabled router for optimal security and minimal latency across the network?

Options:
A. At the Access layer
B. At the Core layer
C. At the Distribution layer
D. It should be distributed across Core, Distribution, and Access layers

✅ **Correct Answer:**
C. At the Distribution layer

Explanation:
In the **Cisco three-tier hierarchical network design (Access, Distribution, Core)**, the **Distribution layer** is responsible for **policy enforcement**, including:

- **Access Control Lists (ACLs)**
- **Routing between VLANs**
- **Security filtering**
- **Quality of Service (QoS) policies**

Placing ACLs at the **Distribution layer** allows policies to be enforced **before traffic reaches the Core**, while keeping the **Core layer fast and optimized for high-speed forwarding**.

Why the others are incorrect:

- **A. Access layer:** Mainly connects end devices (PCs, printers, IP phones).
- **B. Core layer:** Designed for **high-speed packet switching**, not policy enforcement.
- **D. Across all layers:** This would increase complexity and **reduce efficiency**.

7. Which Cisco feature is primarily designed to protect against Layer 2 attacks by verifying the authenticity of DHCP messages?

Options:
A. DHCP Snooping
B. IP Source Guard
C. Port Security
D. Dynamic ARP Inspection (DAI)

✅ **Correct Answer:**
A. DHCP Snooping

Explanation:
DHCP Snooping is a security feature that protects the network from **rogue DHCP servers**. It works by **monitoring and filtering DHCP messages**, allowing only trusted DHCP servers to respond to DHCP requests.

The switch builds a **DHCP snooping binding table**, which maps:

- IP address
- MAC address

- VLAN
- Interface

This table is later used by other security features such as **IP Source Guard** and **Dynamic ARP Inspection (DAI)**.

Why the others are incorrect:

- **B. IP Source Guard:** Prevents **IP spoofing** by filtering traffic based on the DHCP snooping binding table.
- **C. Port Security:** Limits the number of **MAC addresses** on a switch port.
- **D. Dynamic ARP Inspection (DAI):** Protects against **ARP spoofing**, not DHCP attacks.

8. Which of the following DHCP options is used to specify the **TFTP server** to download the bootstrap configuration file?

Options:
A. Option 43
B. Option 6
C. Option 15
D. Option 66

✅ **Correct Answer:**
D. Option 66

Explanation:
DHCP Option 66 is used to specify the **TFTP (Trivial File Transfer Protocol) server name or IP address** from which network devices (such as IP phones, routers, or switches) can download **bootstrap or configuration files**.

Why the others are incorrect:

- **Option 43:** Used for **vendor-specific information** (often used with Cisco wireless controllers or IP phones).
- **Option 6:** Specifies the **DNS server address**.
- **Option 15:** Specifies the **DNS domain name**.

9. Which of the following configurations is necessary to enable DHCP Snooping on a Cisco switch, ensuring that unauthorized DHCP servers are prevented from distributing IP addresses?

Options:
A. Enable the DHCP Snooping feature globally, configure trusted ports, and apply the IP DHCP Snooping for required VLANs.
B. Enable DHCP Snooping globally, configure VLAN filtering, and ensure all ports are set to untrusted by default.
C. Enable the DHCP Snooping feature globally and configure trusted ports.
D. Enable the DHCP Snooping feature locally and disable all DHCP relay functionalities.

✅ **Correct Answer:**
A. Enable the DHCP Snooping feature globally, configure trusted ports, and apply the IP DHCP Snooping for required VLANs.

Explanation:
To properly enable **DHCP Snooping** on a Cisco switch, you must:

1. **Enable DHCP Snooping globally**
2. **Specify which VLANs DHCP Snooping applies to**
3. **Configure trusted ports** (usually the port connected to the legitimate DHCP server)

Example configuration:

```
ip dhcp snooping
ip dhcp snooping vlan 10,20
interface GigabitEthernet0/1
 ip dhcp snooping trust
```

This ensures that **only trusted ports can send DHCP server responses**, preventing **rogue DHCP servers** from assigning incorrect IP addresses.

Why the others are incorrect:

- **B:** DHCP Snooping does not require VLAN filtering configuration.
- **C:** VLAN configuration is also required, not just global enablement.
- **D:** DHCP Snooping is not enabled locally and does not require disabling DHCP relay.

10. In configuring NAT for a small office, which of the following commands correctly configures an inside local address and maps it to an inside global address, and how does it function?

Options:
A. `ip nat inside source list 1 pool NATPOOL`
B. `ip nat outside source static 192.168.1.2 203.0.113.2`
C. `ip nat pool NATPOOL 192.168.1.2 192.168.1.2 netmask 255.255.255.0`
D. `ip nat inside source static 192.168.1.2 203.0.113.2`

✅ **Correct Answer:**
D. `ip nat inside source static 192.168.1.2 203.0.113.2`

Explanation:
The command:

```
ip nat inside source static 192.168.1.2 203.0.113.2
```

creates a **static NAT mapping** between:

- **Inside Local Address:** `192.168.1.2` (private IP inside the network)
- **Inside Global Address:** `203.0.113.2` (public IP seen on the internet)

This means any traffic from the internal device **192.168.1.2** will appear to external networks as **203.0.113.2**. Static NAT creates a **permanent one-to-one mapping** between private and public IP addresses.

Why the others are incorrect:

- **A:** Used for **dynamic NAT with an access list and pool**, not static mapping.
- **B:** `outside source` is used for **outside NAT translations**, not typical inside NAT.
- **C:** Only defines a **NAT pool**, but does not create the translation itself.

11. In an IPv6 network, which type of address is used by multiple devices, but only one device receives the traffic at any given moment?

Options:
A. Multicast
B. Broadcast
C. Unicast
D. Anycast

✅ **Correct Answer:**
D. Anycast

Explanation:
An **Anycast address** in IPv6 is assigned to **multiple devices**, but packets sent to that address are delivered to **only one device**, typically the **nearest or best destination based on routing metrics**.

This is commonly used for:

- **DNS servers**
- **Content delivery networks (CDNs)**
- **Load balancing across servers**

Why the others are incorrect:

- **Multicast:** Sends traffic to **multiple devices simultaneously**.
- **Broadcast:** IPv6 **does not support broadcast**; it uses multicast instead.
- **Unicast:** Sends traffic to **one specific device** only.

12. You have configured a new OSPF network and everything appears to be running correctly. However, when you check the routing table of one of the routers, you notice that it is not receiving any OSPF routes from its neighbors. Which of the following issues is the most likely cause of this problem?

Options:
A. The OSPF hello and dead intervals between routers are mismatched.
B. The autonomous system numbers (ASNs) are mismatched.
C. The OSPF cost is not configured correctly.
D. The OSPF process ID is different on the routers.

✅ **Correct Answer:**
A. The OSPF hello and dead intervals between routers are mismatched.

Explanation:
For **OSPF neighbors to form an adjacency and exchange routes**, several parameters must match, including:

- **Hello interval**
- **Dead interval**

- **Area ID**
- **Authentication (if configured)**
- **Network type**

If the **hello and dead intervals do not match**, the routers will **fail to establish a proper adjacency**, which means **routes will not be exchanged**.

Why the others are incorrect:

- **B. ASN mismatch:** This applies to **BGP**, not OSPF.
- **C. OSPF cost:** Cost only affects **route selection**, not neighbor formation.
- **D. Process ID:** The **OSPF process ID is locally significant**, so it does **not need to match** between routers.

13. Which advanced security feature enables detection and mitigation of network threats by analyzing the behavioral patterns of network traffic?

Options:
A. Virtual Private Network (VPN)
B. Network Access Control (NAC)
C. Next-Generation Intrusion Prevention System (NGIPS)
D. Security Information and Event Management (SIEM)

✅ **Correct Answer:**
C. Next-Generation Intrusion Prevention System (NGIPS)

Explanation:
A **Next-Generation Intrusion Prevention System (NGIPS)** monitors network traffic in real time and uses **deep packet inspection, behavioral analysis, and threat intelligence** to detect and block malicious activities.

NGIPS can:

- Detect **malware and advanced threats**
- Identify **anomalous traffic behavior**
- **Automatically block or mitigate attacks**

Why the others are incorrect:

- **VPN:** Provides **secure encrypted communication**, not threat detection.

- **NAC:** Controls **who or what can access the network**.
- **SIEM:** Collects and analyzes **security logs and events**, but does not directly prevent attacks in real time.

14. Which command is used to configure a DHCP relay agent to forward DHCP client requests to a specific DHCP server with an IP address of **192.168.1.1** on Cisco routers?

Options:
A. `ip helper-address 192.168.1.1`
B. `ip dhcp relay destination 192.168.1.1`
C. `ip dhcp pool 192.168.1.1`
D. `ip dhcp relay information trust-all`

✅ **Correct Answer:**
A. `ip helper-address 192.168.1.1`

Explanation:
The command **`ip helper-address`** is used on a router interface to **forward DHCP broadcast requests from clients to a remote DHCP server** located on another network.

Example configuration:

```
interface GigabitEthernet0/1
 ip helper-address 192.168.1.1
```

This allows DHCP clients on one subnet to obtain IP addresses from a DHCP server on a **different subnet**.

Why the others are incorrect:

- **B:** Not a valid Cisco command.
- **C:** Used to **configure a DHCP pool**, not relay DHCP requests.
- **D:** Related to DHCP relay information options, not forwarding requests.

15. In a network, which of the following statements accurately describes the function of a switch in the context of **Collision Domains and Broadcast Domains**?

Options:
A. A switch maintains both a single collision domain and a single broadcast domain.
B. A switch splits collision domains but maintains a single broadcast domain.

C. A switch maintains a single collision domain but splits broadcast domains.
D. A switch splits both collision domains and broadcast domains.

✅ **Correct Answer:**
B. A switch splits collision domains but maintains a single broadcast domain.

Explanation:
A **Layer 2 switch** creates **separate collision domains for each port**, meaning devices connected to different ports do not experience collisions with each other.

However, by default, all devices connected to the switch are still part of the **same broadcast domain**, meaning broadcast traffic is forwarded to all ports.

Example:

- **Each switch port = separate collision domain**
- **Entire switch = one broadcast domain (unless VLANs are configured)**

Why the others are incorrect:

- **A:** A hub maintains a single collision domain and broadcast domain, not a switch.
- **C:** Switches do not split broadcast domains unless **VLANs or routers** are used.
- **D:** Only **routers or Layer 3 devices** split broadcast domains.

16. Which statement accurately describes how a router processes a packet with a destination address **not in its routing table**, assuming a **default route is set**?

Options:
A. The router forwards the packet based on its default route.
B. The router drops the packet immediately.
C. The router sends an ICMP Redirect message to the source of the packet.
D. The router sends an ARP request to find the MAC address of the destination.

✅ **Correct Answer:**
A. The router forwards the packet based on its default route.

Explanation:
If a router **cannot find a specific route** to a destination in its routing table but **has a default route configured (0.0.0.0/0)**, it forwards the packet to the **next hop specified in the default route**.

Example default route configuration:

```
ip route 0.0.0.0 0.0.0.0 192.168.1.1
```

This means any **unknown destination network** will be sent to **192.168.1.1**.

Why the others are incorrect:

- **B:** The router only drops the packet if **no default route exists**.
- **C:** ICMP Redirect is used when a **better route exists through another gateway**, not when a route is missing.
- **D:** ARP is used to find the **MAC address of the next-hop device**, not the final destination outside the network.

17. Which security mechanism is most effective in protecting against a **zero-day exploit**?

Options:
A. Heuristic Analysis
B. Signature-based Intrusion Detection System (IDS)
C. Patch Management
D. Static Code Analysis

✅ **Correct Answer:**
A. Heuristic Analysis

Explanation:
A **zero-day exploit** targets a vulnerability that is **unknown to the vendor and has no available patch or signature** yet.

Heuristic analysis detects threats by analyzing **behavioral patterns and suspicious activity**, allowing security systems to identify **previously unknown malware or attacks**, including zero-day exploits.

Why the others are incorrect:

- **B. Signature-based IDS:** Detects only **known threats with existing signatures**, so it cannot detect new zero-day attacks effectively.
- **C. Patch Management:** Fixes vulnerabilities **after they are discovered**, but zero-day attacks occur **before patches exist**.
- **D. Static Code Analysis:** Used during **software development** to find vulnerabilities in source code, not to detect live attacks on networks.

18. You have configured a wireless network with **WPA2-Enterprise** security. Despite proper configuration, clients are frequently losing connections and experiencing **high latency**. Which of the following configurations is **MOST likely** to resolve this issue?

Options:

A. Adjust the dynamic frequency selection (DFS) to avoid interference from radar signals.
B. Disable the broadcast of the SSID to reduce network congestion.
C. Increase the idle timeout value on the RADIUS server.
D. Enable Client Load Balancing to evenly distribute the wireless clients.

✅ **Correct Answer:**

D. Enable Client Load Balancing to evenly distribute the wireless clients.

Explanation:

When too many wireless clients connect to a **single access point**, it can cause:

- **High latency**
- **Frequent disconnections**
- **Network congestion**

Client Load Balancing ensures that wireless devices are **distributed across multiple access points**, improving:

- **Performance**
- **Connection stability**
- **Network efficiency**

Why the others are incorrect:

- **A. DFS:** Helps avoid radar interference but is **not the most common cause** of high client latency or disconnections.
- **B. Disable SSID broadcast:** This **does not reduce congestion** and can actually create connection issues.
- **C. Increase RADIUS idle timeout:** This affects **authentication sessions**, not wireless congestion.

19. You have configured OSPF on a network with multiple areas, and you notice that some routes are not being advertised from **Area 0 to Area 1**. You suspect the issue is related to the **OSPF LSA types**. Which LSA type uniquely identifies networks that are originated behind an **Area Border Router (ABR)**?

Options:
A. Type 4 LSA (ASBR Summary LSA)
B. Type 3 LSA (Summary LSA)
C. Type 5 LSA (External LSA)
D. Type 1 LSA (Router LSA)

✅ **Correct Answer:**
B. Type 3 LSA (Summary LSA)

Explanation:
A **Type 3 LSA (Summary LSA)** is generated by an **Area Border Router (ABR)** to advertise **networks from one OSPF area into another area**.

Example scenario:

- Networks in **Area 1** are summarized by the **ABR**
- The ABR sends **Type 3 LSAs** to **Area 0** (and vice versa)

This allows routers in different areas to **learn about networks located in other OSPF areas**.

Why the others are incorrect:

- **Type 1 LSA:** Describes **router links within the same area**.
- **Type 4 LSA:** Advertises the **location of an ASBR**, not the networks themselves.
- **Type 5 LSA:** Used to advertise **external routes redistributed into OSPF** (e.g., from BGP or static routes).

20. You are configuring a **DHCP relay agent** on a Cisco router and need to relay DHCP messages to a DHCP server with the address **192.168.1.1**. Which command should you use to achieve this configuration?

Options:
A. `ip dhcp relay information enable`
B. `ip dhcp-relay server 192.168.1.1`
C. `ip helper-address 192.168.1.1`
D. `dhcp relay address 192.168.1.1`

✅ **Correct Answer:**
C. `ip helper-address 192.168.1.1`

Explanation:
The **`ip helper-address`** command is used on a router interface to **forward DHCP broadcast requests from clients to a DHCP server on a different subnet**.

Example configuration:

```
interface GigabitEthernet0/1
 ip helper-address 192.168.1.1
```

When a DHCP client sends a broadcast request, the router converts it into a **unicast message** and forwards it to the specified DHCP server.

Why the others are incorrect:

- **A.** Enables relay information option but **does not specify the DHCP server**.
- **B.** Not a valid Cisco IOS command.
- **D.** Not a valid Cisco IOS command.

21. You are configuring OSPF on a set of routers in a multi-area network. Which of the following statements about types of LSAs (Link-State Advertisements) and their propagation is correct?

Options:
A. Type 7 LSAs are used in NSSA areas and are converted to Type 5 LSAs by ABRs.
B. Type 5 LSAs are used to propagate external routes within a single OSPF area.
C. Type 3 LSAs are generated by ASBRs to advertise external networks within an area.
D. Type 4 LSAs are used by ABRs to advertise ASBRs to other ABRs in different areas.

✅ **Correct Answer:**
A. Type 7 LSAs are used in NSSA areas and are converted to Type 5 LSAs by ABRs.

Explanation:
In an **OSPF Not-So-Stubby Area (NSSA)**:

- External routes are advertised using **Type 7 LSAs**.
- When these routes reach the **Area Border Router (ABR)**, they are **translated into Type 5 LSAs**.
- The Type 5 LSAs are then propagated to other OSPF areas.

Why the others are incorrect:

- **B:** Type 5 LSAs propagate **external routes across the entire OSPF domain**, not just within a single area.
- **C: Type 3 LSAs** are generated by **ABRs**, not ASBRs.
- **D: Type 4 LSAs** advertise the **location of an ASBR to routers in other areas**, not specifically between ABRs.

22. What is the primary reason for choosing an **OSPF point-to-multipoint network type** over a **non-broadcast multi-access (NBMA)** network type when configuring OSPF in a large-scale WAN environment?

Options:
A. To provide support for dynamic IP addressing in OSPF neighbor configurations.
B. To enhance the scalability of OSPF by increasing the number of OSPF areas.
C. To avoid the need for configuring designated routers (DR) and backup designated routers (BDR) in the network.
D. To reduce the amount of OSPF traffic by consolidating neighbor relationships.

✅ **Correct Answer:**
C. To avoid the need for configuring designated routers (DR) and backup designated routers (BDR) in the network.

Explanation:
The **OSPF point-to-multipoint network type** treats each connection as a **separate point-to-point link**, which means:

- **No DR or BDR election is required**
- Simpler configuration
- Easier neighbor relationships

This is particularly useful in **large WAN environments**, especially with technologies like **Frame Relay or DMVPN**, where full mesh connectivity may not exist.

Why the others are incorrect:

- **A:** OSPF neighbor relationships do **not depend on dynamic IP addressing**.
- **B:** OSPF area scalability is unrelated to the network type.
- **D:** Point-to-multipoint does not consolidate neighbors; it **creates individual logical point-to-point relationships**.

23. Which of the following Ethernet standards supports a **maximum transfer distance of 40 km** while maintaining **full-duplex operation** and a **data rate of 10 Gbps**?

Options:
A. 10GBASE-LR
B. 10GBASE-ER
C. 1000BASE-LX
D. 10GBASE-SR

✅ **Correct Answer:**
B. 10GBASE-ER

Explanation:
10GBASE-ER (Extended Reach) is an Ethernet standard that supports:

- **10 Gbps data rate**
- **Single-mode fiber**
- **Up to 40 km transmission distance**
- **Full-duplex operation**

It is commonly used for **long-distance fiber links**, such as **between buildings or across metropolitan networks**.

Why the others are incorrect:

- **10GBASE-LR:** Supports **10 Gbps up to ~10 km** on single-mode fiber.
- **1000BASE-LX:** Supports **1 Gbps**, not 10 Gbps.
- **10GBASE-SR:** Supports **short distances (~300–400 m)** on multimode fiber.

24. A network administrator notices that a router is **dropping packets during peak hours**. On investigation, it shows that the **input queue is full**.

What steps can the administrator take to alleviate this problem?

Options:
A. Decrease the input queue size and implement traffic shaping.
B. Increase the input queue size and prioritize traffic.
C. Upgrade the router's hardware to increase processing capacity.
D. Apply a Quality of Service (QoS) policy and configure weighted fair queuing.

✅ **Correct Answer:**
C. Upgrade the router's hardware to increase processing capacity.

Explanation:
If the **input queue is consistently full**, it indicates that the router **cannot process incoming packets fast enough**. This usually means the **CPU or hardware processing capability is insufficient** for the traffic load.

Upgrading the router's hardware (or moving to a higher-capacity model) increases:

- **Packet processing speed**
- **Buffer capacity**
- **Overall throughput**

This is the most effective long-term solution when the router is **hardware-limited**.

Why the others are incorrect:

- **A:** Decreasing the queue size would **increase packet drops**, not solve the issue.
- **B:** Increasing queue size may temporarily buffer packets but **does not fix the processing bottleneck**.
- **D:** QoS can help prioritize traffic but **does not solve a fundamental hardware processing limitation**.

25. In a Cisco router, how do you enable **Network Address Translation (NAT)** to translate traffic from an internal private IP address range **(192.168.1.0/24)** to a **single public IP address (203.0.113.1)** for all outgoing traffic?

Options:
A. Use the command `ip nat inside source list 1 interface Serial0/0 overload`
B. Use the command `ip nat inside source list 1 pool PUBLIC_IP_POOL overload`
C. Use the command `ip nat inside source list 1 interface Serial0/0`
D. Use the command `ip nat inside source list 1 pool NAT_POOL overload`

✅ **Correct Answer:**
A. `ip nat inside source list 1 interface Serial0/0 overload`

Explanation:
This command enables **PAT (Port Address Translation)**, allowing **multiple internal private IP addresses** to share **one public IP address** assigned to the router interface.

Example configuration:

```
access-list 1 permit 192.168.1.0 0.0.0.255
ip nat inside source list 1 interface Serial0/0 overload
```

How it works:

- **Access list 1** defines the internal network (`192.168.1.0/24`)
- The router uses the **IP address of Serial0/0** as the public address
- **Overload** allows many internal devices to share that **single public IP using different ports**

Why the others are incorrect:

- **B & D:** These require a **configured NAT pool**, which is unnecessary when using a **single interface IP**.
- **C:** Missing the **overload keyword**, so it would not enable PAT for multiple hosts.

26. In a network, you need to configure an **Access Control List (ACL)** to **block all traffic from the range 192.168.10.0/24 except for traffic to one specific IP address, 192.168.10.5**. Which of the following **standard ACL statements** would achieve this?

Options:
A. `access-list 1 permit any`
`access-list 1 deny 192.168.10.5 0.0.0.0`

B. `access-list 1 permit 192.168.10.0 0.0.0.255`

C. `access-list 1 deny 192.168.10.0 0.0.0.255`

D. `access-list 1 permit 192.168.10.5 0.0.0.0`
`access-list 1 deny 192.168.10.0 0.0.0.255`

✅ **Correct Answer:**
D.

```
access-list 1 permit 192.168.10.5 0.0.0.0
access-list 1 deny 192.168.10.0 0.0.0.255
```

Explanation:
ACLs are processed **top-down**, meaning the router evaluates rules in order.

Steps:

1. **Permit traffic from 192.168.10.5**
2. **Deny all other traffic from the 192.168.10.0/24 network**

Because of the **implicit "deny any" at the end of every ACL**, all other traffic will also be blocked.

Why the others are incorrect:

- **A:** Would allow all traffic because of `permit any`.
- **B:** Would **allow the entire subnet**, not block it.
- **C:** Would **deny the entire subnet**, including `.5`.

27. Which of the following statements about **IPv6 addressing** is correct?

Options:
A. IPv6 supports broadcast addresses to efficiently send traffic to all nodes on a network.
B. In an IPv6 address, triple colons `:::` can be used to represent a sequence of multiple 16-bit groups of zeros.
C. An IPv6 address can contain both upper-case and lower-case letters without being case-sensitive.
D. IPv6 addresses are 128-bit long and are represented in hexadecimal format.

✅ **Correct Answer:**
D. IPv6 addresses are 128-bit long and are represented in hexadecimal format.

Explanation:
An **IPv6 address** is **128 bits long** and written as **eight groups of four hexadecimal digits**, separated by colons.

Example:

```
2001:0db8:85a3:0000:0000:8a2e:0370:7334
```

Key points:

- Uses **hexadecimal format (0–9, A–F)**
- Each group represents **16 bits**
- Total = **128 bits**

Why the others are incorrect:

- **A:** IPv6 **does not support broadcast**; it uses **multicast instead**.
- **B:** IPv6 allows only **double colon (::)** once in an address, not **:::**.
- **C:** IPv6 is **not case-sensitive**, but the more fundamental correct statement is **that IPv6 addresses are 128-bit hexadecimal**.

28. In a network using **IPv6**, which type of address is used for **one-to-nearest communication**, typically in scenarios involving services like **DHCPv6**?

Options:
A. Broadcast Address
B. Unicast Address
C. Anycast Address
D. Multicast Address

✅ **Correct Answer:**
C. Anycast Address

Explanation:
An **Anycast address** is assigned to **multiple devices**, but when traffic is sent to that address, it is delivered to **only one device**, typically the **nearest or best destination according to the routing protocol**.

This is commonly used for services such as:

- **DNS servers**
- **DHCP services**
- **Content delivery networks (CDNs)**

The benefit is **efficient routing and load distribution**, as clients automatically reach the **closest available server**.

Why the others are incorrect:

- **Broadcast Address:** IPv6 **does not support broadcast**.
- **Unicast Address:** Communication is **one-to-one**, not one-to-nearest.
- **Multicast Address:** Sends packets to **multiple devices simultaneously**, not just the nearest one.

29. Which command should be used to configure the **SNMP agent to send logging data to a specific remote host**?

Options:
A. `snmp-server host`
B. `snmp-server enable informs`
C. `snmp-server enable traps`
D. `logging trap`

✅ **Correct Answer:**
A. `snmp-server host`

Explanation:
The **`snmp-server host`** command specifies the **remote SNMP management station (host)** that will receive **SNMP traps or informs** from the Cisco device.

Example:

```
snmp-server host 192.168.1.10 public
```

This command configures the router or switch to **send SNMP notifications to the specified remote host**.

Why the others are incorrect:

- **B. `snmp-server enable informs`** – Enables SNMP **inform messages**, but does not specify the destination host.
- **C. `snmp-server enable traps`** – Enables traps globally but **does not define where to send them**.
- **D. `logging trap`** – Configures **Syslog severity level**, not SNMP.

30. When configuring **port security** on a switch, which command must be used to **limit the number of allowed MAC addresses to a single specific MAC address on a port in sticky mode**, without initially learning any MAC addresses?

Options:
A. `switchport port-security maximum 1`
B. `switchport port-security mac-address sticky`
C. `switchport port-security mac-address sticky mac_address`
D. `switchport port-security mac-address mac_address`

✅ **Correct Answer:**
C. `switchport port-security mac-address sticky mac_address`

Explanation:
The command **`switchport port-security mac-address sticky mac_address`** allows you to **manually configure a specific MAC address while still using sticky learning mode**.

Sticky MAC addresses are **dynamically learned and then stored in the running configuration**, but you can also **define a specific MAC address directly** using this command.

Example:

```
interface FastEthernet0/1
 switchport mode access
 switchport port-security
 switchport port-security maximum 1
 switchport port-security mac-address sticky 00AA.BBCC.DDEE
```

Why the others are incorrect:

- **A.** Only limits the number of MAC addresses but **does not specify which MAC address is allowed**.
- **B.** Enables sticky learning but **does not define a specific MAC address**.
- **D.** Configures a **static secure MAC address**, but **not in sticky mode**.

31. You have configured an **EtherChannel** on a Cisco switch using **LACP mode**. However, the EtherChannel is not forming and shows a **"suspended"** state. What could be a possible reason for this issue?

Options:
A. The configured number of links exceeds the limit set for the EtherChannel.
B. Inconsistent port configurations, such as mismatched VLANs, on the EtherChannel

member ports.
C. The EtherChannel is configured in manual ON mode on one end.
D. Different LACP system priorities on both ends.

✅ **Correct Answer:**
B. Inconsistent port configurations, such as mismatched VLANs, on the EtherChannel member ports.

Explanation:
For an **EtherChannel to form correctly**, all member interfaces must have **identical configurations**, including:

- VLAN settings
- Trunk/access mode
- Speed and duplex
- Allowed VLANs
- Native VLAN

If these settings **do not match**, the switch may place the interface in a **"suspended" state**, preventing the EtherChannel from forming.

Why the others are incorrect:

- **A:** EtherChannel usually supports **up to 8 active links**, but exceeding this limit is not the common cause of the suspended state.
- **C:** If one side is set to **ON** and the other uses **LACP**, the channel typically **fails to negotiate**, not necessarily shows suspended due to mismatch configs.
- **D:** Different **LACP system priorities** do not prevent EtherChannel formation; they only influence which device controls the negotiation.

32. In the context of implementing **VLANs** in a network, which of the following advanced configurations would allow for **dynamic VLAN assignments without requiring manual configuration on each switch port**?

Options:
A. Access ports with PVID assignments
B. Trunk ports with 802.1Q encapsulation
C. Dynamic VLANs with the use of VMPS
D. Static VLANs with port membership

✅ **Correct Answer:**
C. Dynamic VLANs with the use of VMPS

Explanation:
VMPS (VLAN Membership Policy Server) allows switches to **dynamically assign VLANs based on the MAC address of the connected device**.

How it works:

- A device connects to a switch port.
- The switch sends the **MAC address to the VMPS server**.
- The VMPS server determines the **appropriate VLAN**.
- The switch **automatically assigns the VLAN** to that port.

This eliminates the need to **manually configure VLAN membership on each port**.

Why the others are incorrect:

- **A. Access ports with PVID:** VLAN must still be **manually configured**.
- **B. Trunk ports with 802.1Q:** Used to **carry multiple VLANs between switches**, not dynamic assignment.
- **D. Static VLANs:** Require **manual configuration of each port**.

33. In a wireless network, what is the purpose of the **802.11r** standard?

Options:
A. Improves quality of service by prioritizing traffic types.
B. Supports fast roaming by allowing clients to quickly transition between access points.
C. Increases the bandwidth available for wireless communications.
D. Enhances security by implementing stronger encryption techniques.

✅ **Correct Answer:**
B. Supports fast roaming by allowing clients to quickly transition between access points.

Explanation:
The **IEEE 802.11r** standard, also known as **Fast BSS Transition (FT)**, is designed to improve **roaming performance in wireless networks**.

Key benefits:

- Allows **devices to move quickly between access points (APs)**
- Reduces **authentication delay during roaming**
- Important for **real-time applications** such as:
    - VoIP
    - Video calls
    - Real-time collaboration

Without **802.11r**, a device must **fully reauthenticate with each AP**, which can cause **latency or dropped connections**.

Why the others are incorrect:

- **A:** QoS prioritization is handled by **802.11e (WMM)**.
- **C:** Bandwidth improvements come from newer standards like **802.11ac or 802.11ax**.
- **D:** Stronger encryption is provided by **WPA2/WPA3**, not 802.11r.

34. You are configuring **DHCP Snooping** on a Cisco switch. Which of the following steps is **NOT necessary** to correctly implement DHCP snooping?

Options:
A. Enable DHCP Option 82.
B. Designate trusted and untrusted interfaces.
C. Configure trusted DHCP servers on the switch.
D. Enable DHCP snooping for the required VLANs.

✅ **Correct Answer:**
A. Enable DHCP Option 82.

Explanation:
DHCP Option 82 (Relay Agent Information Option) is **optional** in DHCP Snooping configurations. It adds additional information such as the **switch port and VLAN** to DHCP messages, which can help DHCP servers make better address assignments, but it is **not required for DHCP Snooping to function**.

To properly configure DHCP Snooping, the necessary steps are:

1. **Enable DHCP Snooping globally**
2. **Enable DHCP Snooping on the required VLANs**
3. **Configure trusted ports** (usually the port connected to the DHCP server)

Example configuration:

```
ip dhcp snooping
ip dhcp snooping vlan 10,20
interface GigabitEthernet0/1
 ip dhcp snooping trust
```

Why the others are required:

- **B:** Trusted ports must be configured so the switch knows which ports can send DHCP responses.
- **C:** The switch must allow communication with legitimate DHCP servers through trusted ports.
- **D:** DHCP Snooping must be enabled on specific VLANs where DHCP traffic exists.

35. Which of the following statements about the **Spanning Tree Protocol (STP)** and its role in Ethernet networks is **FALSE**?

Options:
A. STP operates at the network layer (Layer 3) of the OSI model.
B. STP is used to prevent loops in a network by creating a loop-free tree structure.
C. STP ensures that there is a single active path between any two network devices.
D. A Root Bridge is elected based on the lowest Bridge ID in an STP network.

❌ **False Statement (Correct Answer):**
A. STP operates at the network layer (Layer 3) of the OSI model.

Explanation:
Spanning Tree Protocol (STP) operates at the **Data Link Layer (Layer 2)** of the OSI model, not at Layer 3.

STP works by:

- **Preventing Layer 2 loops**
- **Blocking redundant paths**
- **Creating a loop-free logical topology**

Why the others are correct:

- **B:** STP prevents loops by building a **spanning tree topology**.

- **C:** STP ensures **only one active path** between devices while keeping backup paths blocked.
- **D:** The **Root Bridge** is selected based on the **lowest Bridge ID (Bridge Priority + MAC Address)**.

36. Which of the following **security mechanisms** can be used to prevent an attacker from forwarding **malicious ARP packets** to other devices in a network?

Options:
A. Dynamic ARP Inspection (DAI)
B. DHCP Snooping
C. IP Source Guard
D. Port Security

✅ **Correct Answer:**
A. Dynamic ARP Inspection (DAI)

Explanation:
Dynamic ARP Inspection (DAI) protects the network from **ARP spoofing/ARP poisoning attacks**. It works by **intercepting and validating ARP packets** on the network.

DAI verifies ARP packets against the **DHCP Snooping binding table**, which contains:

- IP address
- MAC address
- VLAN
- Switch port

If an ARP packet does **not match the binding table**, it is **dropped**, preventing attackers from sending **fake ARP responses**.

Why the others are incorrect:

- **B. DHCP Snooping:** Protects against **rogue DHCP servers**, not ARP attacks.
- **C. IP Source Guard:** Prevents **IP address spoofing**, not specifically ARP poisoning.
- **D. Port Security:** Limits **MAC addresses per port**, but does not validate ARP packets.

37. What is the primary difference between **802.1X** and **port security** when applied to **wired network access**?

Options:
A. Port security provides encryption for data transmitted over wired networks, while 802.1X does not.
B. Port security provides a centralized authentication mechanism, while 802.1X uses MAC address filtering.
C. 802.1X enforces a maximum of three MAC addresses per port, while port security uses digital certificates.
D. 802.1X provides a centralized authentication mechanism, while port security uses MAC address filtering.

✅ **Correct Answer:**
D. 802.1X provides a centralized authentication mechanism, while port security uses MAC address filtering.

Explanation:
802.1X is a **network access control protocol** that uses a **central authentication server (typically RADIUS)** to verify user or device credentials before granting access to the network.

Port Security, on the other hand, works by:

- **Allowing specific MAC addresses** on a switch port
- **Limiting the number of MAC addresses**
- Blocking unknown or unauthorized devices

Key difference:

Feature	802.1X	Port Security
Authentication	Centralized (RADIUS)	Local MAC filtering
Security Level	Higher	Basic
User/Device Validation	Username/password, certificates	MAC address only

Why the others are incorrect:

- **A:** Port security does **not provide encryption**.
- **B:** The roles are reversed.
- **C:** Port security controls the **number of MAC addresses**, not 802.1X.

38. Which of the following **Ethernet frame fields** uniquely identifies the **upper-layer protocol** encapsulated in the frame?

Options:
A. Source MAC Address
B. Preamble
C. EtherType
D. Frame Check Sequence (FCS)

✅ **Correct Answer:**
C. EtherType

Explanation:
The **EtherType field** in an Ethernet frame identifies which **upper-layer protocol** is being carried in the payload.

Examples of EtherType values:

EtherType	Protocol
0x0800	IPv4
0x86DD	IPv6
0x0806	ARP

This allows the receiving device to determine **how to process the encapsulated data**.

Why the others are incorrect:

- **Source MAC Address:** Identifies the **sender device**, not the protocol.
- **Preamble:** Used for **synchronization** between devices.
- **FCS (Frame Check Sequence):** Used for **error detection**, not protocol identification.

39. In a corporate network, a security engineer needs to ensure that **all device communication is authenticated and encrypted using digital certificates**. Which of the following protocols should be implemented to achieve this objective?

Options:
A. 802.1X
B. SSL/TLS

C. SSH
D. IPsec

✅ **Correct Answer:**
D. IPsec

Explanation:
IPsec (Internet Protocol Security) is designed to **secure IP communications by authenticating and encrypting every IP packet** in a network session. It can use **digital certificates** for authentication and provides **end-to-end encryption**.

Key features of IPsec:

- **Authentication Header (AH)** – Ensures data integrity and authentication
- **Encapsulating Security Payload (ESP)** – Provides **encryption and confidentiality**
- **Certificate-based authentication** via PKI

IPsec is commonly used for:

- **Site-to-site VPNs**
- **Remote access VPNs**
- **Secure network communication**

Why the others are incorrect:

- **802.1X:** Used for **network access control**, not encrypting all traffic.
- **SSL/TLS:** Secures **application-layer communication** (e.g., HTTPS), not all network traffic.
- **SSH:** Secures **remote login sessions**, not general network communication.

40. Which of the following configuration steps is necessary to **enable 802.1X authentication on a Cisco switch port**?

Options:
A. Configure the switch port as an access port and enable dot1x.
B. Configure VLAN membership on the port.
C. Configure an IP address on the switch.
D. Enable port security on the interface.

✅ **Correct Answer:**
A. Configure the switch port as an access port and enable dot1x.

Explanation:
To enable **802.1X authentication** on a Cisco switch port, the port must be configured as an **access port**, and **802.1X authentication must be enabled**.

Example configuration:

```
interface FastEthernet0/1
 switchport mode access
 authentication port-control auto
 dot1x pae authenticator
```

This allows the switch to act as an **authenticator**, verifying devices through a **RADIUS server** before granting network access.

Why the others are incorrect:

- **B:** VLAN configuration is optional and **not required to enable 802.1X**.
- **C:** An IP address on the switch is **not required for port-level 802.1X authentication**.
- **D: Port security** is a different mechanism that uses **MAC address filtering**, not centralized authentication.

41. Which **IPv6 address type** is designed to be used **within a single site**, **never to be routed on the global internet**, and allows **unique addressing for local communications within a specific network**?

Options:
A. Link-local Address
B. Multicast Address
C. Global Unicast Address
D. Unique Local Address (ULA)

✅ **Correct Answer:**
D. Unique Local Address (ULA)

Explanation:
A **Unique Local Address (ULA)** is an IPv6 address used for **private internal networks**, similar to **private IPv4 addresses** (like 192.168.x.x).

Key characteristics:

- Range: **fc00::/7** (commonly **fd00::/8**)
- **Not routable on the global internet**
- Used for **internal communication within an organization or site**

Example:

```
fd12:3456:789a::1
```

Why the others are incorrect:

- **Link-local (fe80::/10):** Used only for **local link communication**, not across an entire site.
- **Multicast:** Used to send traffic to **multiple devices simultaneously**.
- **Global Unicast:** Routable **across the internet**, similar to public IPv4 addresses.

42. Which of the following **port security violation modes** will cause the interface to become **error-disabled and shut down immediately** if a violation occurs?

Options:
A. Shutdown
B. Protect
C. Shutdown VLAN
D. Restrict

✅ **Correct Answer:**
A. Shutdown

Explanation:
The **Shutdown** violation mode is the **default port security action** on Cisco switches. When a violation occurs:

- The interface is **immediately placed in an error-disabled state**
- The port is **shut down**
- A **log message and SNMP trap** are generated

Example configuration:

```
interface FastEthernet0/1
 switchport port-security
 switchport port-security violation shutdown
```

Why the others are incorrect:

- **Protect:** Drops packets from unauthorized MAC addresses but **does not shut down the port**.
- **Restrict:** Drops unauthorized packets and **generates logs**, but the port **remains active**.
- **Shutdown VLAN:** Not a valid Cisco port security violation mode.

43. A network administrator needs to **prioritize voice traffic over data traffic** in a large enterprise network. To achieve this, the administrator configured **policy-based routing (PBR)**. However, users are experiencing **increased latency in voice calls**. Which of the following actions can resolve this issue?

Options:
A. Increase the bandwidth for the entire network.
B. Implement Quality of Service (QoS) mechanisms to prioritize voice traffic.
C. Use a higher routing metric for data traffic.
D. Enable IPv6 for better traffic handling.

✅ **Correct Answer:**
B. Implement Quality of Service (QoS) mechanisms to prioritize voice traffic.

Explanation:
Quality of Service (QoS) is specifically designed to **prioritize delay-sensitive traffic**, such as **voice and video**, over normal data traffic.

QoS techniques include:

- **Traffic classification**
- **Traffic marking (DSCP/CoS)**
- **Priority queuing**
- **Bandwidth reservation**

For example, **VoIP traffic can be marked with DSCP EF (Expedited Forwarding)** so it is processed with **higher priority and minimal delay**.

Why the others are incorrect:

- **A:** Increasing bandwidth may help but **does not guarantee priority for voice traffic**.
- **C:** Routing metrics affect **path selection**, not traffic prioritization.
- **D:** IPv6 does **not directly reduce voice latency**.

44. In a Cisco switch network, which command would you use to **mitigate a VLAN hopping attack on a trunk port**?

Options:
A. `switchport trunk encapsulation dot1q`
B. `switchport mode dynamic desirable`
C. `switchport mode access`
D. `switchport nonegotiate`

✅ **Correct Answer:**
D. `switchport nonegotiate`

Explanation:
A **VLAN hopping attack** can occur when an attacker exploits **Dynamic Trunking Protocol (DTP)** to automatically negotiate a trunk link.

The command:

```
switchport nonegotiate
```

Disables DTP, preventing the port from automatically negotiating a trunk connection. This helps **reduce the risk of VLAN hopping attacks**.

Example configuration:

```
interface GigabitEthernet0/1
 switchport mode trunk
 switchport nonegotiate
```

Why the others are incorrect:

- **A. `switchport trunk encapsulation dot1q`:** Specifies the trunk encapsulation but **does not prevent VLAN hopping**.
- **B. `switchport mode dynamic desirable`:** Actually **enables DTP negotiation**, increasing the risk.
- **C. `switchport mode access`:** Used for access ports, not trunk ports.

45. You have configured an **NTP server** on Router A and several **multicast clients**. However, the clients are not synchronizing their clocks correctly. Upon checking, you realized the issue is related to the **multicast address configuration**. Which multicast address is appropriate for **NTP service** and why?

Options:
A. 225.0.0.1
B. 224.0.1.1
C. 224.0.0.5
D. 239.255.255.250

✅ **Correct Answer:**
B. 224.0.1.1

Explanation:
The multicast address **224.0.1.1** is reserved for **Network Time Protocol (NTP)** multicast synchronization.

Key points:

- Used by **NTP servers to send time updates to multiple clients simultaneously**
- Allows **efficient time synchronization** without individual unicast queries
- Belongs to the **globally scoped multicast address range**

Why the others are incorrect:

- **225.0.0.1:** Not reserved for NTP services.
- **224.0.0.5:** Used by **OSPF routers** for routing updates.
- **239.255.255.250:** Used by **SSDP/UPnP service discovery**, not NTP.

46. In an **OSPF network configuration**, which of the following statements about the **Designated Router (DR)** and **Backup Designated Router (BDR)** is correct?

Options:
A. The DR/BDR election only occurs within OSPF area 0, the backbone area.
B. The BDR directly forwards OSPF updates to all routers in case the DR fails, but it does not form adjacencies with other routers.
C. The DR and BDR must always be manually configured on each OSPF router.
D. The DR communicates with all other routers on the OSPF network segment directly, while other routers communicate with each other through the DR and BDR.

✅ **Correct Answer:**
D. The DR communicates with all other routers on the OSPF network segment directly, while other routers communicate with each other through the DR and BDR.

Explanation:
In **multi-access networks** (like Ethernet), OSPF elects a **Designated Router (DR)** and **Backup Designated Router (BDR)** to reduce the number of adjacencies and routing updates.

How it works:

- All routers form **full adjacency with the DR and BDR**
- Routers send updates to the **DR**
- The **DR distributes updates to other routers**
- The **BDR acts as a standby** and takes over if the DR fails

This improves **scalability and reduces routing overhead**.

Why the others are incorrect:

- **A:** DR/BDR elections occur on **any multi-access network**, not only Area 0.
- **B:** The BDR **does form adjacencies** with other routers.
- **C:** DR/BDR election is **automatic**, though priority can influence the election.

47. When troubleshooting a **route-based VPN** in a network, you find that there is **asymmetric routing** occurring, which causes the tunnel to fail. Which technique can be used to resolve this issue?

Options:
A. Configuring Equal-Cost Multi-Path (ECMP)
B. Using a Route-Based VPN with Reverse Route Injection (RRI)

C. Implementing Policy-Based Routing (PBR)
D. Enabling Dead Peer Detection (DPD)

✅ **Correct Answer:**
B. Using a Route-Based VPN with Reverse Route Injection (RRI)

Explanation:
Reverse Route Injection (RRI) automatically installs routes into the routing table for networks reachable through the **VPN tunnel**. This helps ensure that return traffic follows the **same path through the VPN**, preventing **asymmetric routing**.

Benefits of RRI:

- Maintains **symmetric routing paths**
- Automatically **updates routing tables**
- Helps ensure traffic is correctly routed through the **VPN tunnel**

Why the others are incorrect:

- **A. ECMP:** Used for **load balancing across multiple equal-cost paths**, not specifically to fix asymmetric VPN routing.
- **C. PBR:** Can influence routing decisions but is **not the standard solution for route-based VPN asymmetric routing**.
- **D. DPD:** Detects **dead VPN peers**, but does not resolve routing path issues.

48. You are tasked with configuring a **small office network**. The current setup includes a router and several switches. You need to decide the **best IP addressing scheme** to optimize the use of IP addresses while ensuring that each department has its own subnet. Which IP addressing scheme would you choose and why?

Options:
A. Allocate a **/24 subnet** to each department
B. Allocate a **/30 subnet** to each department
C. Use a **single /24 subnet** for the entire network
D. Use **Variable Length Subnet Masking (VLSM)** to allocate appropriate subnet sizes to each department

✅ **Correct Answer:**
D. Use Variable Length Subnet Masking (VLSM) to allocate appropriate subnet sizes to each department

Explanation:
VLSM allows you to create **different subnet sizes based on the actual needs of each department**. This helps:

- **Conserve IP addresses**
- **Avoid waste**
- **Provide separate subnets for each department**
- **Improve network organization and scalability**

Example:

- HR may need **/28**
- Finance may need **/27**
- IT may need **/26**

Instead of giving every department the same subnet size, **VLSM matches subnet size to real host requirements**.

Why the others are incorrect:

- **A. /24 for each department:** Works, but wastes many IP addresses in a small office.
- **B. /30 for each department:** Too small; only supports **2 usable host addresses**.
- **C. Single /24 for entire network:** Does not separate departments into different subnets.

49. Which protocol is primarily responsible for **browser-based operations** and supports methods like **GET** and **POST** in **application layer communication**?

Options:
A. DNS
B. FTP
C. IMAP
D. HTTP

✅ **Correct Answer:**
D. HTTP

Explanation:
HTTP (HyperText Transfer Protocol) is the primary **application-layer protocol used for web communication** between browsers and web servers.

Common HTTP methods include:

- **GET** – Requests data from a server (e.g., loading a webpage)
- **POST** – Sends data to the server (e.g., submitting a form)
- **PUT** – Updates resources
- **DELETE** – Removes resources

Example:

```
GET /index.html HTTP/1.1
Host: example.com
```

Why the others are incorrect:

- **DNS:** Resolves **domain names to IP addresses**.
- **FTP:** Used for **file transfers**.
- **IMAP:** Used for **email retrieval** from mail servers.